

ANIMAL INTRUSION FOMO MODEL

Independent V3 Evaluation Report

AI / IoT Based Smart Farm Perimeter Protection System

Evaluation Summary	
Test Images	50
True Positive	29
True Negative	19
False Positive	1
False Negative	1
Accuracy	96.00%
Precision	96.67%
Recall	96.67%
F1 Score	96.67%
False Positive Rate	5.00%
Average Confidence	0.8305
Average Latency	6.10 ms

Confusion Matrix

	Predicted Intrusion	Predicted Non-Intrusion
Actual Intrusion	29 (TP)	1 (FN)
Actual Non-Intrusion	1 (FP)	19 (TN)

Model	with cats and cow.eim
Input resolution	160 × 160 RGB
Test set	30 positive + 20 negative
Positive class	Cattle + dogs + cats = animal intrusion
Decision threshold	0.50

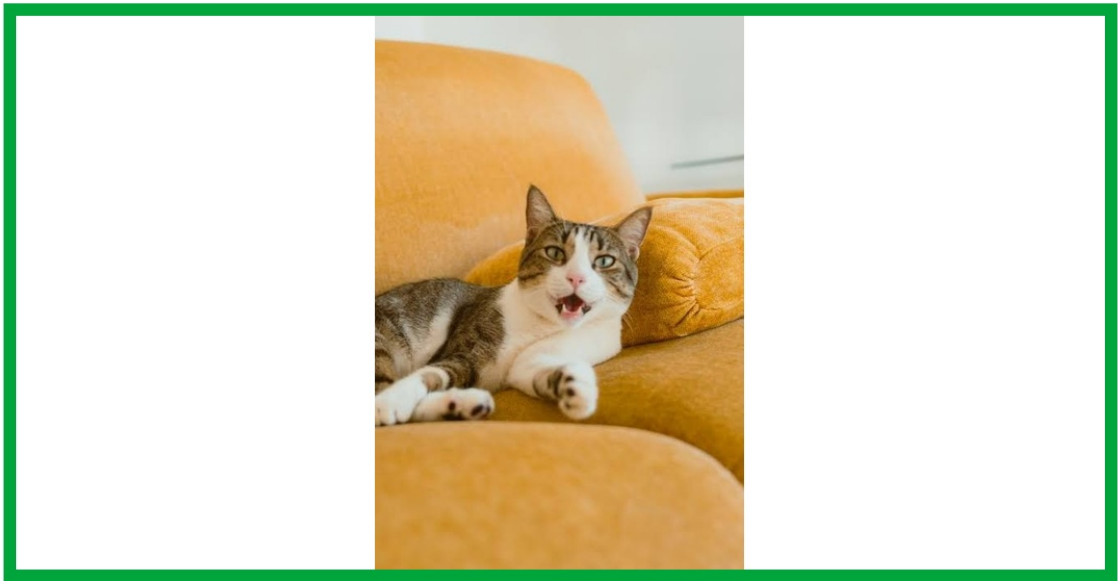
MODEL DEVELOPMENT HISTORY

- Cleaned cattle dataset: 1,271 images.
- Training split: 1,016 images.
- Validation split: 255 images.
- FOMO MobileNetV2 0.35.
- 60 training cycles.
- Learning rate: 0.001.
- CPU training.
- Data augmentation enabled.
- INT8 quantized model.
- 350 greenery/farm-background negative images were added.
- Independent testing identified white/light cattle as an important false-negative pattern.
- White/light cattle examples were added.
- Dogs and cats were incorporated into the positive animal-intrusion class.
- V3 test set: 15 cattle + 8 dogs + 7 cats + 20 non-cattle.
- Final accuracy: 96.00%.
- Final precision: 96.67%.
- Final recall: 96.67%.
- Final F1 score: 96.67%.
- Final false-positive rate: 5.00%.

Positive subtype	Correct	Total	Recall
Cattle	15	15	100.00%
Dog	7	8	87.50%
Cat	7	7	100.00%

cattle_001.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 85.16%
Cattle Boxes: 3
Latency: 11.670 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 85.16% | Boxes: 3 | Latency: 11.670 ms

cattle_002.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 76.95%
Cattle Boxes: 3
Latency: 6.690 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 76.95% | Boxes: 3 | Latency: 6.690 ms

cattle_003.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 94.92%
Cattle Boxes: 2
Latency: 6.420 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 94.92% | Boxes: 2 | Latency: 6.420 ms

cattle_004.jpg

CORRECT



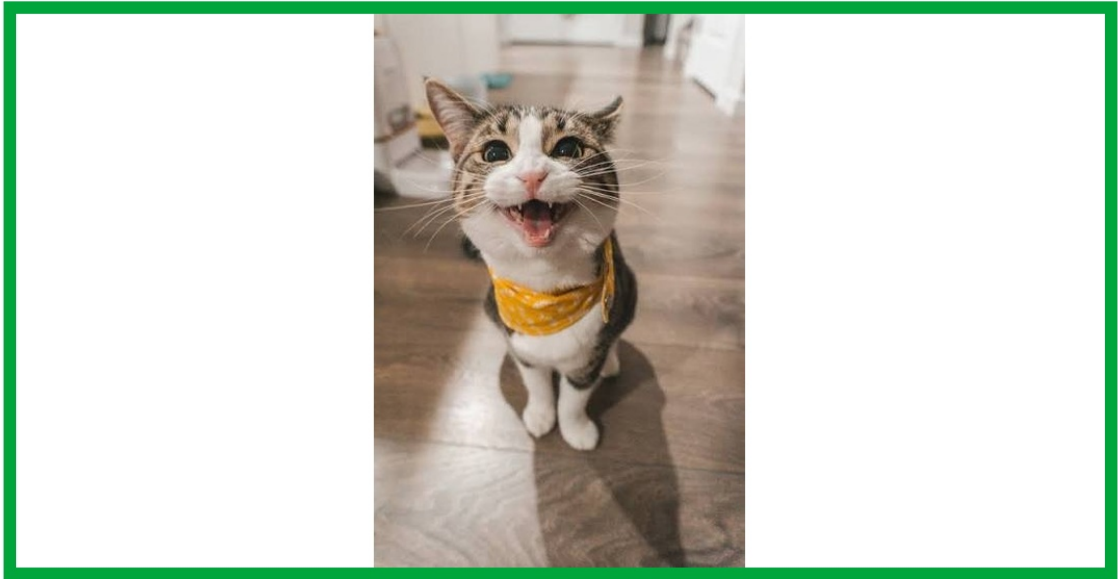
Ground Truth: cattle
Prediction: cattle
Confidence: 90.62%
Cattle Boxes: 6
Latency: 6.950 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 90.62% | Boxes: 6 | Latency: 6.950 ms

cattle_005.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 76.95%
Cattle Boxes: 2
Latency: 6.850 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 76.95% | Boxes: 2 | Latency: 6.850 ms

cattle_006.jpg

CORRECT



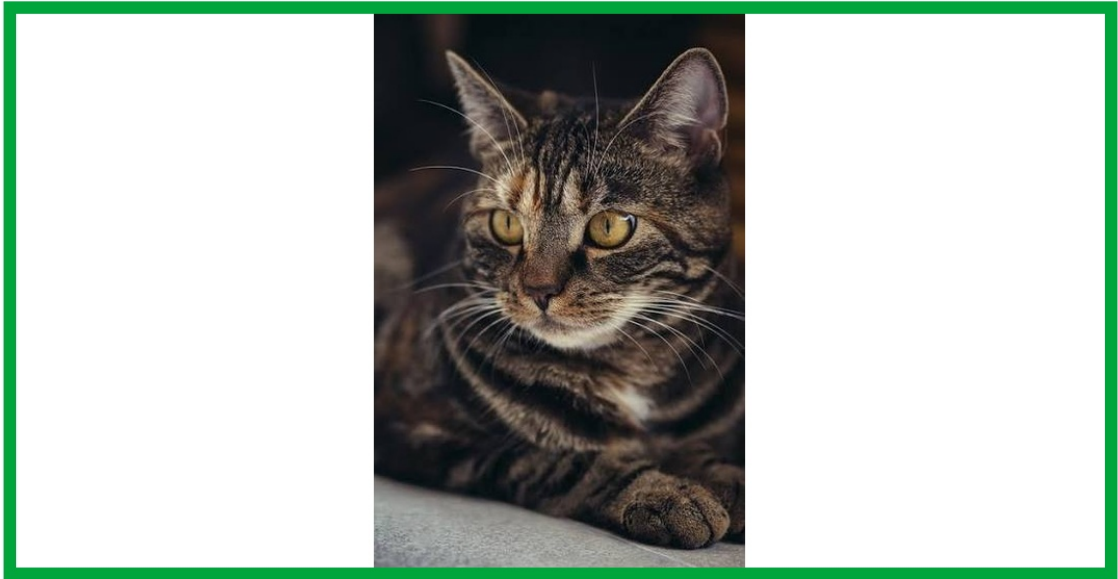
Ground Truth: cattle
Prediction: cattle
Confidence: 90.62%
Cattle Boxes: 4
Latency: 6.330 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 90.62% | Boxes: 4 | Latency: 6.330 ms

cattle_007.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 90.62%
Cattle Boxes: 4
Latency: 6.260 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 90.62% | Boxes: 4 | Latency: 6.260 ms

cattle_008.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 92.97%
Cattle Boxes: 4
Latency: 5.830 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 92.97% | Boxes: 4 | Latency: 5.830 ms

cattle_009.jpg

CORRECT



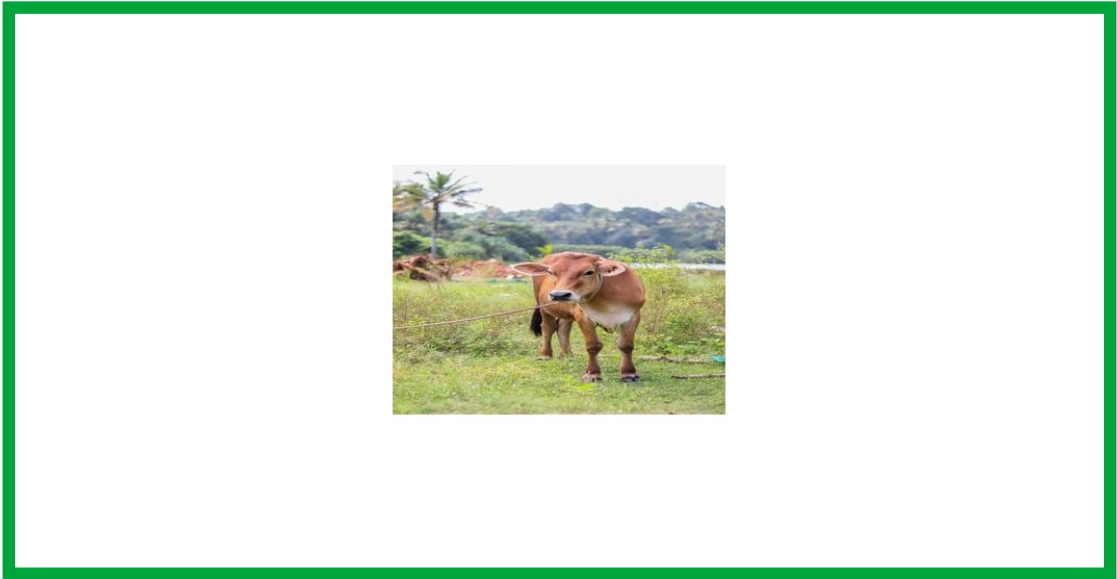
Ground Truth: cattle
Prediction: cattle
Confidence: 85.16%
Cattle Boxes: 2
Latency: 6.110 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 85.16% | Boxes: 2 | Latency: 6.110 ms

cattle_010.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 89.06%
Cattle Boxes: 2
Latency: 5.540 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 89.06% | Boxes: 2 | Latency: 5.540 ms

cattle_011.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 94.92%
Cattle Boxes: 4
Latency: 5.820 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 94.92% | Boxes: 4 | Latency: 5.820 ms

cattle_012.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 62.89%
Cattle Boxes: 1
Latency: 5.390 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 62.89% | Boxes: 1 | Latency: 5.390 ms

cattle_013.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 98.44%
Cattle Boxes: 6
Latency: 5.740 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 98.44% | Boxes: 6 | Latency: 5.740 ms

cattle_014.jpg**CORRECT**

Ground Truth: cattle
Prediction: cattle
Confidence: 82.81%
Cattle Boxes: 2
Latency: 5.710 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 82.81% | Boxes: 2 | Latency: 5.710 ms

cattle_015.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 92.97%
Cattle Boxes: 3
Latency: 5.650 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 92.97% | Boxes: 3 | Latency: 5.650 ms

cattle_016.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 90.62%
Cattle Boxes: 3
Latency: 6.040 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 90.62% | Boxes: 3 | Latency: 6.040 ms

cattle_017.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 50.00%
Cattle Boxes: 2
Latency: 5.400 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 50.00% | Boxes: 2 | Latency: 5.400 ms

cattle_018.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 96.88%
Cattle Boxes: 4
Latency: 6.260 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 96.88% | Boxes: 4 | Latency: 6.260 ms

cattle_019.jpg**CORRECT**

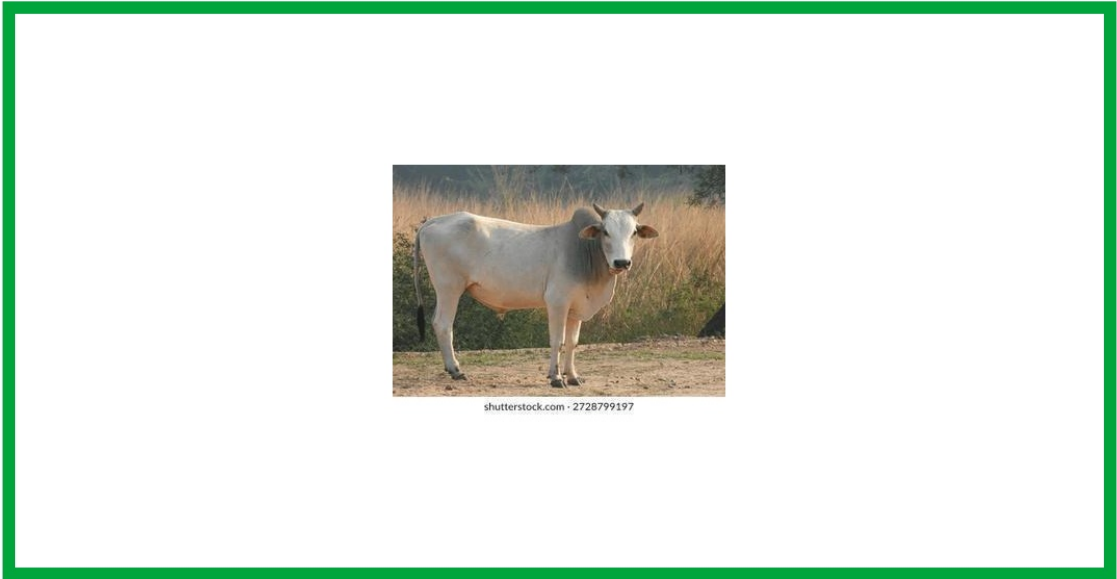
Ground Truth: cattle
Prediction: cattle
Confidence: 82.81%
Cattle Boxes: 3
Latency: 5.440 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 82.81% | Boxes: 3 | Latency: 5.440 ms

cattle_020.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 94.92%
Cattle Boxes: 4
Latency: 5.960 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 94.92% | Boxes: 4 | Latency: 5.960 ms

cattle_021.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 70.31%
Cattle Boxes: 3
Latency: 5.800 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 70.31% | Boxes: 3 | Latency: 5.800 ms

cattle_022.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 73.83%
Cattle Boxes: 2
Latency: 5.740 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 73.83% | Boxes: 2 | Latency: 5.740 ms

cattle_023.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 82.81%
Cattle Boxes: 6
Latency: 6.550 ms
Running Accuracy100.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 82.81% | Boxes: 6 | Latency: 6.550 ms

cattle_024.jpg

WRONG

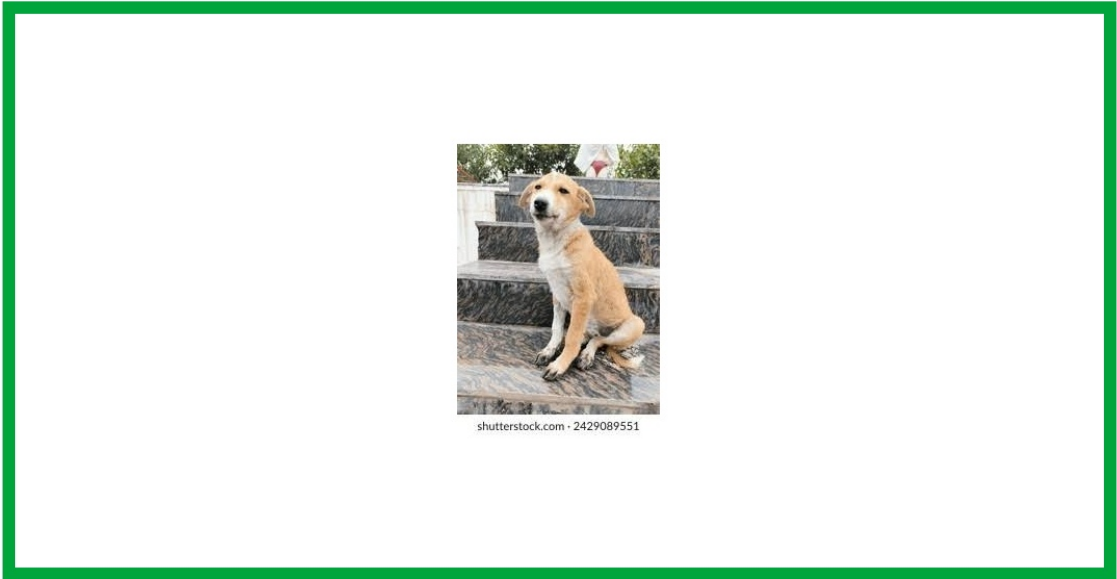
Ground Truth: cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.720 ms
Running Accuracy95.83%

IMAGE RESULT: WRONG

GT: cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.720 ms

cattle_025.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 70.31%
Cattle Boxes: 4
Latency: 5.700 ms
Running Accuracy96.00%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 70.31% | Boxes: 4 | Latency: 5.700 ms

cattle_026.jpg

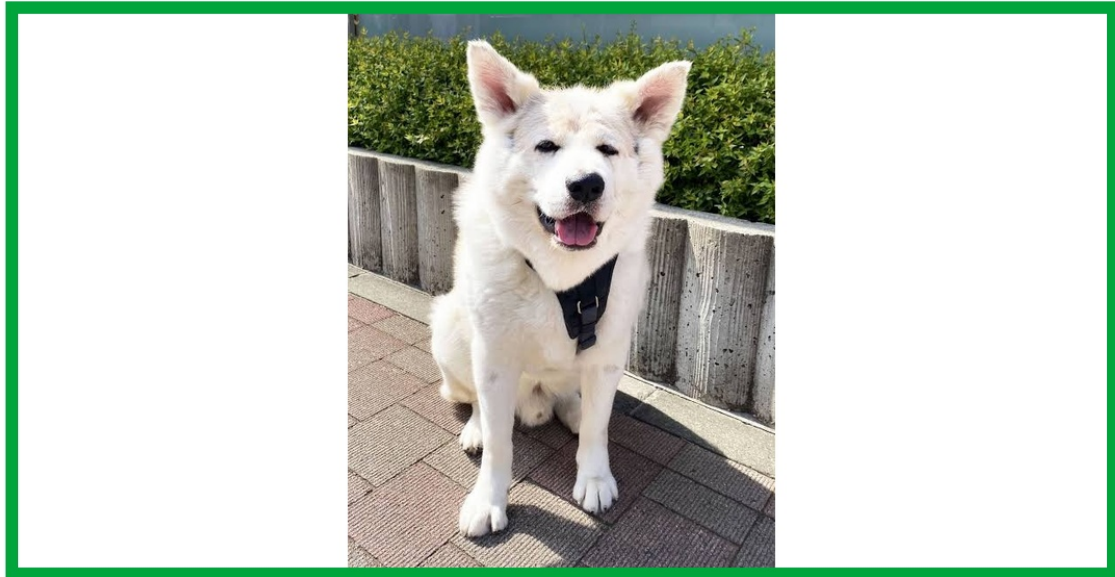
CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 85.16%
Cattle Boxes: 2
Latency: 6.230 ms
Running Accuracy96.15%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 85.16% | Boxes: 2 | Latency: 6.230 ms

cattle_027.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 73.83%
Cattle Boxes: 3
Latency: 6.310 ms
Running Accuracy96.30%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 73.83% | Boxes: 3 | Latency: 6.310 ms

cattle_028.jpg

CORRECT

Ground Truth: cattle
Prediction: cattle
Confidence: 98.05%
Cattle Boxes: 2
Latency: 10.150 ms
Running Accuracy96.43%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 98.05% | Boxes: 2 | Latency: 10.150 ms

cattle_029.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 70.31%
Cattle Boxes: 2
Latency: 6.000 ms
Running Accuracy96.55%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 70.31% | Boxes: 2 | Latency: 6.000 ms

cattle_030.jpg

CORRECT



Ground Truth: cattle
Prediction: cattle
Confidence: 96.48%
Cattle Boxes: 2
Latency: 6.060 ms
Running Accuracy96.67%

IMAGE RESULT: CORRECT

GT: cattle | Prediction: cattle | Confidence: 96.48% | Boxes: 2 | Latency: 6.060 ms

noncattle_001.jpg

CORRECT



Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 5.710 ms

Running Accuracy96.77%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.710 ms

noncattle_002.jpg

CORRECT



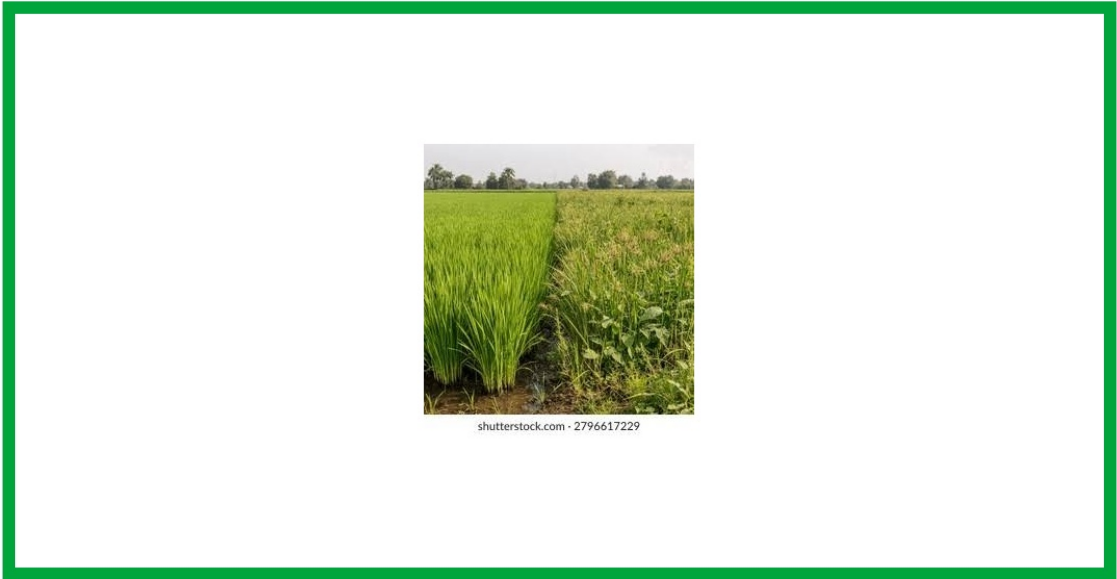
Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.360 ms
Running Accuracy96.88%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.360 ms

noncattle_003.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.830 ms
Running Accuracy96.97%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.830 ms

noncattle_004.jpg

CORRECT



Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 5.440 ms

Running Accuracy97.06%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.440 ms

noncattle_005.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.720 ms
Running Accuracy97.14%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.720 ms

noncattle_006.jpg

CORRECT**Ground Truth:** non_cattle**Prediction:** non_cattle**Confidence:** 0.00%**Cattle Boxes:** 0**Latency:** 6.240 ms**Running Accuracy**97.22%**IMAGE RESULT: CORRECT**

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 6.240 ms

noncattle_007.jpg

CORRECT



Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 5.300 ms

Running Accuracy97.30%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.300 ms

noncattle_008.jpg

CORRECT



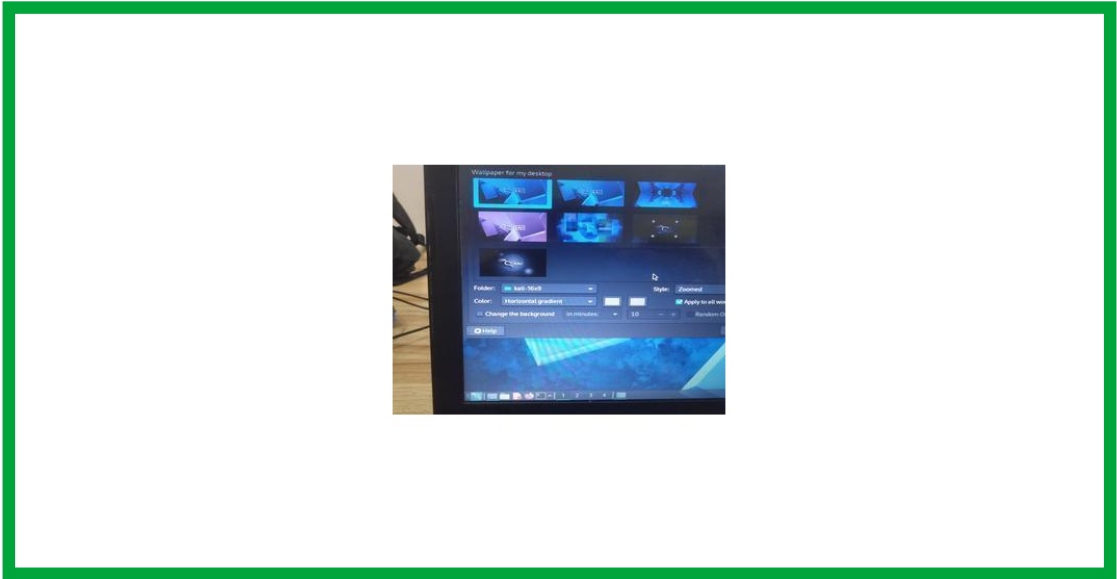
Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.160 ms
Running Accuracy97.37%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.160 ms

noncattle_009.jpg

CORRECT



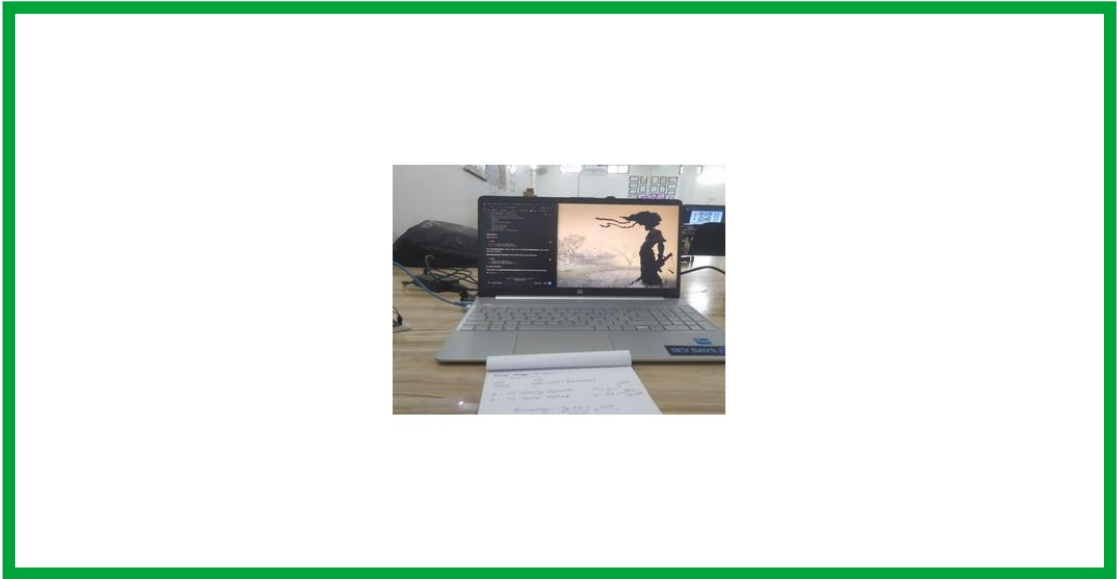
Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.390 ms
Running Accuracy97.44%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.390 ms

noncattle_010.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.890 ms
Running Accuracy97.50%

IMAGE RESULT: CORRECT

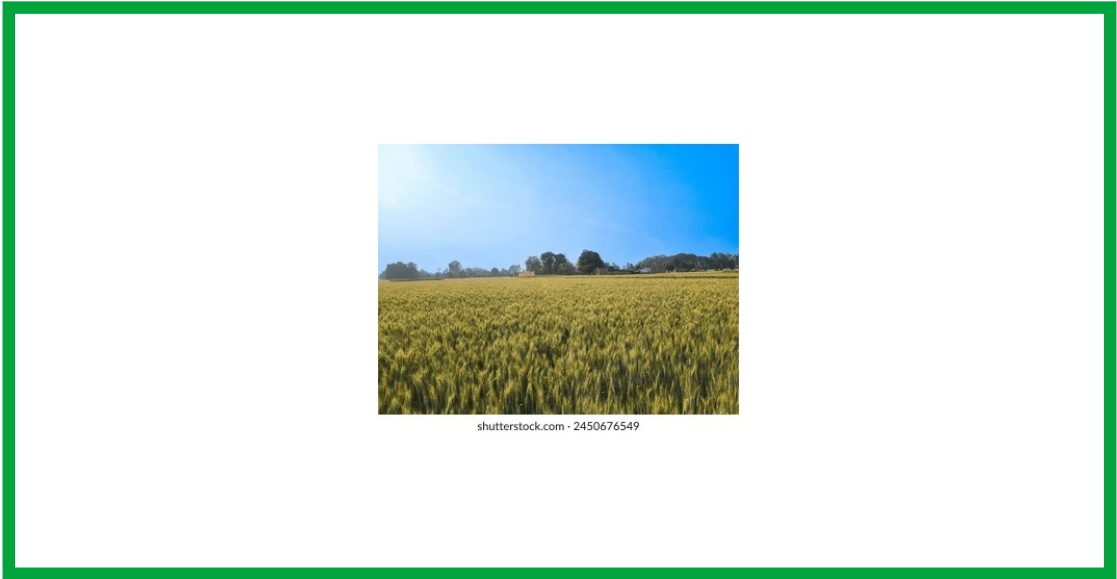
GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.890 ms

noncattle_011.jpg**CORRECT****Ground Truth:** non_cattle**Prediction:** non_cattle**Confidence:** 0.00%**Cattle Boxes:** 0**Latency:** 6.440 ms**Running Accuracy**97.56%**IMAGE RESULT: CORRECT**

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 6.440 ms

noncattle_012.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 6.380 ms
Running Accuracy97.62%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 6.380 ms

noncattle_013.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.260 ms
Running Accuracy97.67%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.260 ms

noncattle_014.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 5.880 ms
Running Accuracy97.73%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.880 ms

noncattle_015.jpg

CORRECT



Ground Truth: non_cattle

Prediction: non_cattle

Confidence: 0.00%

Cattle Boxes: 0

Latency: 5.450 ms

Running Accuracy97.78%

IMAGE RESULT: CORRECT

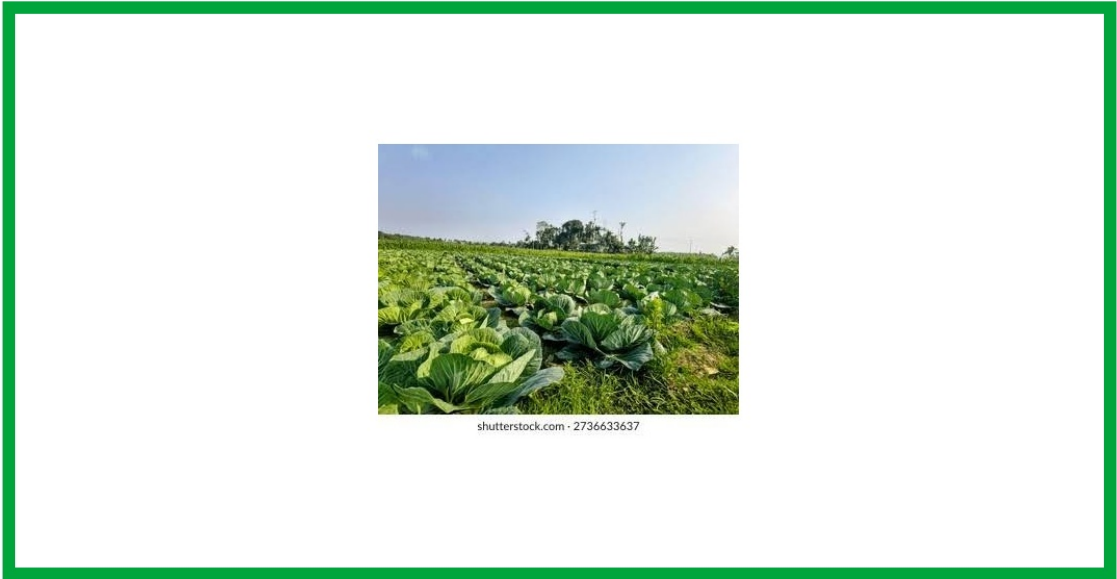
GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.450 ms

noncattle_016.jpg**CORRECT****Ground Truth:** non_cattle**Prediction:** non_cattle**Confidence:** 0.00%**Cattle Boxes:** 0**Latency:** 5.740 ms**Running Accuracy**97.83%**IMAGE RESULT: CORRECT**

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.740 ms

noncattle_017.jpg

CORRECT



Ground Truth: non_cattle
Prediction: non_cattle
Confidence: 0.00%
Cattle Boxes: 0
Latency: 6.030 ms
Running Accuracy97.87%

IMAGE RESULT: CORRECT

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 6.030 ms

noncattle_018.jpg

WRONG



Ground Truth: non_cattle

Prediction: cattle

Confidence: 50.00%

Cattle Boxes: 1

Latency: 5.830 ms

Running Accuracy95.83%

IMAGE RESULT: WRONG

GT: non_cattle | Prediction: cattle | Confidence: 50.00% | Boxes: 1 | Latency: 5.830 ms

noncattle_019.jpg**CORRECT****Ground Truth:** non_cattle**Prediction:** non_cattle**Confidence:** 0.00%**Cattle Boxes:** 0**Latency:** 5.530 ms**Running Accuracy**95.92%**IMAGE RESULT: CORRECT**

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.530 ms

noncattle_020.jpg**CORRECT****Ground Truth:** non_cattle**Prediction:** non_cattle**Confidence:** 0.00%**Cattle Boxes:** 0**Latency:** 5.910 ms**Running Accuracy**96.00%**IMAGE RESULT: CORRECT**

GT: non_cattle | Prediction: non_cattle | Confidence: 0.00% | Boxes: 0 | Latency: 5.910 ms